

Geospatial flow data extraction and visualization using Earth mover's distance

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Abstract

A flow map is a common visualization technique displaying the significant trends and changes in data over a given spatial and temporal domain. While most flow maps are constructed using trajectory datasets as input, only a few research initiatives focus on producing flow maps from other data forms. This paper uses the **Earth Mover's Distance to extract flows from geospatial data** in two consecutive temporal snapshots. We further perform a generalization of the flows to avoid visual cluttering. To demonstrate the proposed approach, we provide an interactive web application where the user can see the pre-calculated flows extracted from the 2021 COVID incidence data in the regions of Germany, Poland, and Czechia, accessible at <https://adammertel.github.io/flow-extraction/>.

Methodology

Earth Movers distance (EMD)

EMD is a metric that **compares two mass distributions** by calculating the distance representing the minimal costs needed to transform one distribution into another. The inputs to the EMD estimation are the source and target distributions and a **cost matrix** representing the cost of moving one mass unit between each source-target node combination. EMD is widely applicable for image comparison, but researchers have successfully applied the method in the geospatial context as well, where it helped to visualize, analyze, or validate data in fields like urban research, land management, and animal movement.

Geospatial flow data extraction

- Data Preparation**
 - rescaling data and reducing precision
 - preparing cost matrix
- EMD estimation**
 - selecting EDM method and assigning estimation parameters
 - flow extracting in a form of arrows
- Flow Generalization**
 - grouping arrows based on their proximity (graph based approach)
 - calculating generalized arrows

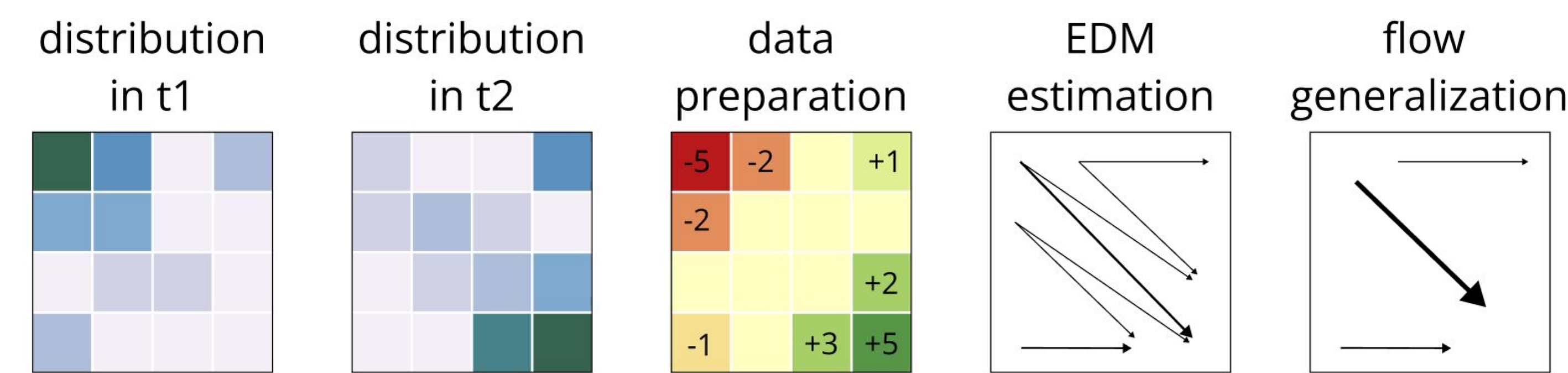


Fig 1. Overview of the flow extraction and visualization steps

Results

Demonstration

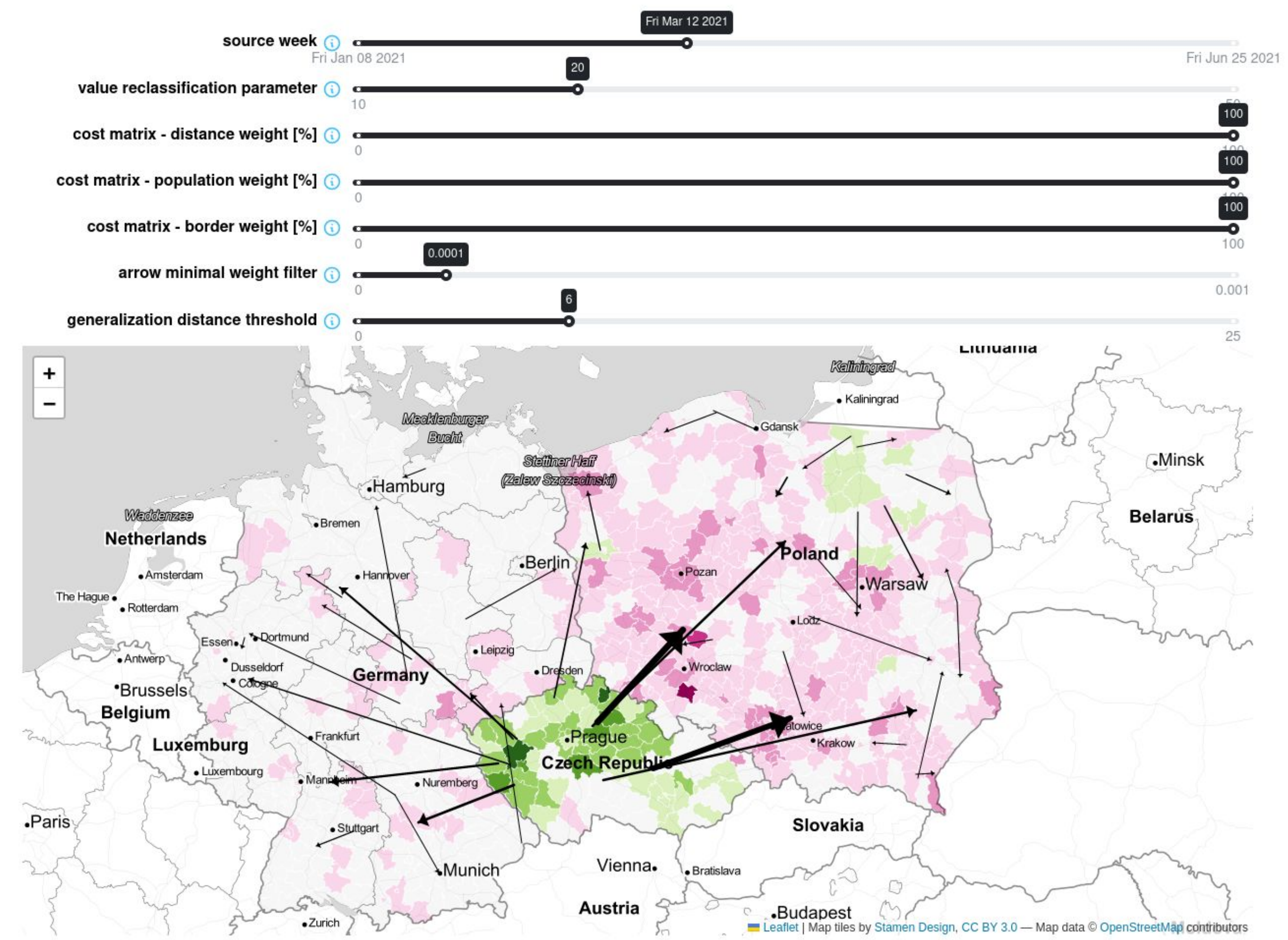
To demonstrate our approach, we developed an interactive web application extracting flows from a dataset of COVID-19 incidence in the regions of Germany, Poland, and the Czech Republic in 2021 based on the parameter values of the user's input.



demonstration application

Geospatial flow data extraction

Flow data extraction from the weekly COVID-19 incidence in values of the regions of Germany, Poland, and Czechia. The calculation is based on the [Earth mover's distance](#) method. Version 0.2.1 (Aug 22, 2022)



Potential advantages of EMD based flow extraction

Automatic Flow extraction also from non-movement data ("a metaphor of movement")

Generalized arrows offer graphically non-extensive visualization - a possibility to combine with other geovisualization methods

The EDM-based methodology offers a "geometrically meaningful" data metric

Customizability and extendability through various EMD estimation techniques, their extensions, and setting parameters to fit the geographical context of the presented topic and the use of the outcome

Future work

- research the usability of various EDM methods, application of constraints and additional rules focusing on the geospatial context
- Wasserstein Barycenters and EDM-based interpolation
- improvement of the generalization process
- testing the usability of the visualization technique

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